



Research progress of zero-shot learning

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Abstract

Although there have been encouraging breakthroughs in supervised learning since the renaissance of deep learning, the recognition of large-scale object classes remains a challenge, especially when some classes have no or few training samples. In this paper, the development of ZSL is reviewed comprehensively, including the evolution, key technologies, mainstream models, current research hotspots and future research directions. First, the evolution process is introduced from the perspectives of multi-shot, few-shot to zero-shot learning. Second, the key techniques of ZSL are analyzed in detail in terms of three aspects: visual feature extraction, semantic representation and visual-semantic mapping. Third, some typical models are interpreted in chronological order. Finally, closely related articles from the last three years are collected to analyze the current research hotspots and list future research directions.

Keywords Zero-shot learning · Feature extraction · Semantic representation · Visual-semantic mapping

1 Introduction

Traditional visual recognition technologies usually require a large number of labeled training data, such as thousands of images or even tens of thousands of images for each class, to achieve good classification accuracy. However, it is impossible to collect such a large amount of data with labels for all classes, and this is a huge challenge for traditional supervised learning. At present, there are some different solutions, for example, semi-supervised learning [1], transfer learning [2], self-taught learning [3], and few-shot learning [4]. It has been found that the fewer training samples there are, the more difficult recognition is; when there are no training samples, the process is called zero-shot learning (ZSL) [5].

Human beings have the ability to retain or accumulate acquired knowledge from past learning tasks and quickly integrate this knowledge to solve new recognition tasks; this provides inspiration for researchers to attempt to endow machines with the same intelligence, for which the ability of continuous

learning/lifelong learning [6] is needed. In this process, ZSL is of great significance for the realization of the ultimate goal of “intelligence”.

ZSL can be regarded as a special type of cross-modal retrieval learning. It is a method for the recognition of new classes using unlabeled data, and the basic idea is to transfer knowledge from seen classes to unseen classes by sharing attributes. Most existing ZSL methods use an intermediate semantic representation (such as visual features or semantic word vectors), and the semantic representation is shared between the labeled auxiliary data set and the unlabeled target data set. In particular, the projection from the low-level feature space to the semantic space is learned from the auxiliary data set, which does not need to be adaptive to the target data set.

In recent years, researchers have performed much work with regard to ZSL; for example, Fu et al. [7] proposed a multiview embedding framework to solve the problem of projection domain shift. For the same problem, Zhao et al. [8] combined two spaces into a joint embedded space and provided a comprehensive refinement mechanism to learn the shared subspace structure. It is important to synthesize missing data in the joint embedded space, as in similar studies [9]. Qin et al. [10] adopted a new method of learning semantic-driven attributes. This method can directly learn the optimal binary attributes by effectively solving discrete optimization problems. Zhang et al. [11] proposed a framework for data-related feature transfer learning that was based on the optimization of the parameter family of local feature displacement to maximize the source-target adaptive similarity function. Guo et al. [12]

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developed a new end-to-end recognition framework, which completes recognition tasks in the original feature space; therefore, it can significantly reduce redundancy. Ye et al. [13] studied a method of progressive integrated network modeling based on multi-projection labeled embedding. Wang et al. [14] proposed a solution of adopting the nonlinear compatibility function between the feature space sample and the semantic space prototype. However, in the latest study [15], an unsupervised clustering technique was combined with ZSL to solve the problem of multivariate data classification. The zero-shot classifier is very effective in cases with a new class containing zero training samples, for which a cluster-based zero-shot learning algorithm [16] is applied to deal with seriously unbalanced data.

The ZSL aims to eliminate the constraint of labeled data on artificial intelligence systems. The study of ZSL not only has great theoretical value for the whole field of artificial intelligence but also has tremendous application potential. With the maturity of deep learning technology, the research landscape of ZSL has been reshaped in almost all aspects, such as algorithm designs, training/testing data sets, application scenarios and even evaluation protocols. Therefore, it is important to review zero-shot learning. Although there have been some review works on ZSL [17] or related methods, most of these studies summarized and compared techniques of specific branches of ZSL, such as in reviews of zero-shot action recognition [18], few-shot learning [19], and open set recognition [20]. Unfortunately, as these were early publications, they were unable to cover the latest developments of a highly-systematic ZSL approach; therefore, our work is of great significance.

The rest of this article is organized as follows. In Section 2, the evolution of the ZSL framework is discussed. In Section 3, according to the implementation steps of ZSL, the key techniques are expounded. In Section 4, the current typical ZSL models are introduced in chronological order, and in Section 5, as a further extension of ZSL, the latest development in the study of generalized ZSL is discussed. In Section 6, research on ZSL in the past three years is obtained, and then hotspots and future challenges are listed. Finally, a conclusion is drawn in Section 7.

2 Evolution of network architecture

ZSL, which was defined as zero-data learning, was first proposed in the field of artificial intelligence in 2008 [21]. It is used to solve the classification problems with insufficient labeled training data. In 2009, Mark et al. [22] extended such a problem to the field of neural information processing and named it “zero-shot learning” for the first time. In the same year, Lampert et al. [23] extended the problem to the field of computer vision and suggested using semantic attributes such

as the shapes, colors and even local information of objects to help recognize unseen classes.

ZSL has experienced a long development process. Unsupervised learning and supervised learning are two cores in the field of machine learning. The former is conducted by using a large amount of unlabeled data to train the classifier and then using the intrinsic relationships between samples to perform classification. The latter is carried out using a large number of labeled data to train the classifier, and then complete classification relies on the relationship between each feature and its label. Semi-supervised learning is a method that combines unsupervised and supervised learning. It uses a small amount of labeled data and a large amount of unlabeled data to train the classifier, and this significantly improves its recognition performance for seen classes. The difference between few-shot learning and semi-supervised learning is the number of samples, with few-shot learning requiring fewer than 10 labeled samples to complete the classification process. However, ZSL is the most extreme case, where the test classes are absolutely unseen. For generalized ZSL (GZSL), test samples include unseen classes and seen classes. Details on the evolution of ZSL can be seen in Table 1.

Table 1 shows that ZSL is different from previous supervised learning methods because the unseen testing set and seen training set are mutually exclusive. Nevertheless, ZSL can recognize the unseen class by establishing a relationship between the seen class and the unseen class, as shown in Fig. 1.

As seen in Fig. 1, generally, this relationship is built through an embedding space, which is the core of ZSL. First, feature vectors from the original image are extracted to construct the feature space, and then semantic vectors from the corresponding class labels are extracted to construct the semantic space. Finally, the mapping relationships between the spaces are analyzed to construct the embedding space. Based on the semantic space, the relation between feature and class labels is learned by using the seen class data set during the training stage; furthermore, the relation is used to complete the prediction process in the testing stage. First, a semantic vector is predicted by the corresponding image features, and then the category of the image is matched according to the semantic vector. Equation (1) is as follows.

$$\begin{aligned} C &= I(f(x)) \\ f &: x \rightarrow k \\ l &: k \rightarrow z \end{aligned} \quad (1)$$

where “ I ” is obtained from the training data set and used to complete the mapping of an image feature “ x ” to the embedding space “ k ”. Equation (1) generally adopts the nearest-neighbors algorithm or another more complex similarity calculation method to predict the category label “ z ” of the image.

At present, existing ZSL methods can be divided into two types according to the use of unlabeled data in the training stage: the transductive type and the inductive type.

Table 1 Evolution process of the ZSL method

Type	Training set y	Testing set z	"R" relationship of y and z	Classifier "C"
Unsupervised learning	Large amount of unlabeled images	Seen class	$R: y = z$	$C: x \rightarrow y$
Supervised learning	Large amount of labeled images	Seen class	$R: y = z$	$C: x \rightarrow y$
Semi-supervised learning	a small amount of labeled and a large amount of unlabeled images	Seen class	$R: y = z$	$C: x \rightarrow y$
Few-shot learning	Few labeled and a large amount of unlabeled images	Seen class	$R: y = z$	$C: x \rightarrow y$
ZSL	Large amount of labeled images and attributes	Unseen class	$R: y \cap z = \emptyset$	$C: x \rightarrow z$
GZSL	Large amount of labeled images and attributes	Seen class and unseen class	$R: y \cap z = y$	$C: x \rightarrow z$

Note: "x" represents the feature space of the images, "y" stands for the training classes, and "z" denotes the test classes. When "y" and "z" are mutually exclusive, " $x \rightarrow y$ " represents a classifier built by x to implement a prediction for a seen class, and " $x \rightarrow z$ " represents a classifier built by x to conduct a prediction for unseen classes or seen+unseen classes

2.1 Transductive type

Transductive ZSL means that unlabeled data are available during the training stage of the classifier, as shown in Fig. 2 (a). On the one hand, this strategy can improve the generalization ability of the model on the unseen class; on the other hand, the data of unseen and seen classes can be combined to jointly optimize the classifier. The motivation of this method is that the visual space X and semantic embedding space K have their own manifold structures, and both manifold structure alignments are as far as possible during the process of model learning. Then, the mapping function learned in this case exhibits improved transfer capability on the test set. Transductive-type ZSL was first proposed by in 2012 by Fu et al. [24], who introduced a generative model for user-defined attributes. The attribute distribution of the test data is predicted by the average k-nearest neighbors (KNN) of the prototype. Another representative work of transductive ZSL is [25],

which discussed metrics for analyzing the relationship between the manifold structure and the transfer ability of the model for different semantic embedding spaces and then constructed an optimization model for the manifold structure. Verma et al. [26] also used the generative framework, which can naturally represent each class using a probability distribution. Their method also used the unlabeled data from unseen classes and employed transfer/semi-supervised learning to improve the estimation of the class distribution. It is worth noting that their method can be easily self-updated by providing a small number of additional labels, and it can then be easily extended to few-shot learning [27].

2.2 Inductive type

Inductive-type ZSL refers to the fact that only the training set used in the training phase of the classifier, which then predicts the class labels in parallel during the testing phase, as shown in

Fig. 1 Zero-shot learning diagram

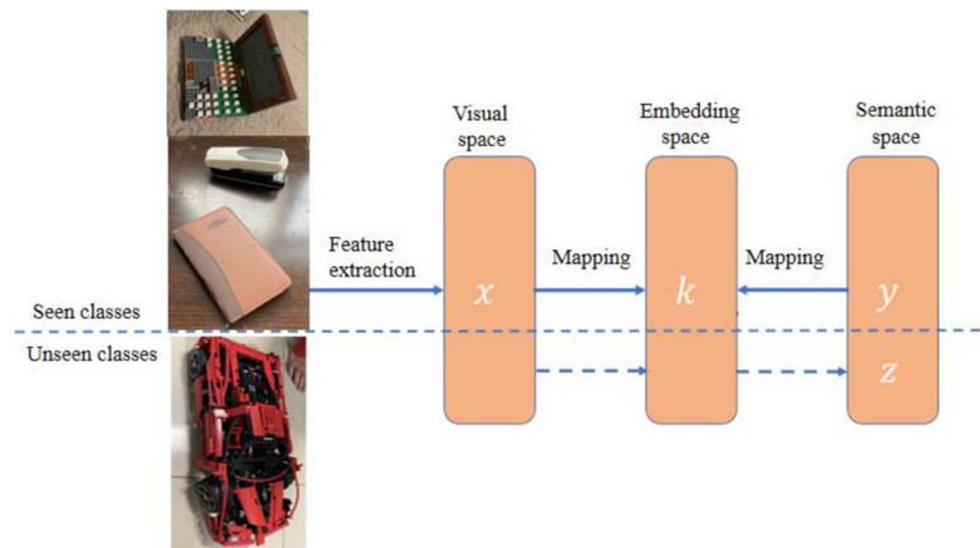


Fig. 2 Schematic diagram of ZSL. (a) Transductive type; (b) Inductive type

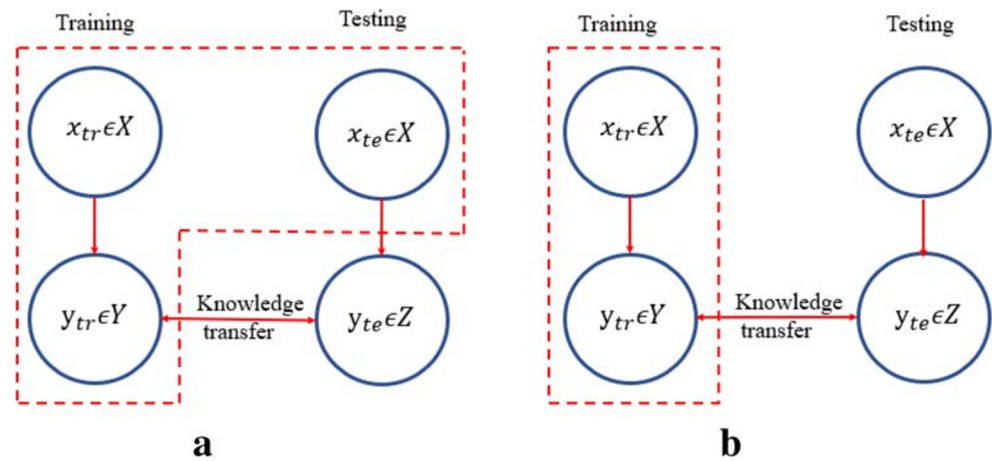


Fig. 2(b). Training samples in the transductive-type ZSL model have stringent requirements for unlabeled test samples, and this limits the ZSL training process to some extent. However, inductive-type ZSL has relatively good flexibility and extensibility, so it is in line with the requirements of practical problems. Therefore, it is also currently the most widely used ZSL method.

3 Key techniques for ZSL

The general steps used by researchers for ZSL problems are visual feature extraction, semantic representation, and visual-semantic mapping, as shown in Fig. 3.

3.1 Visual feature extraction

In the field of computer vision, the extraction of visual features is very important, and how to effectively extract the most valuable features has been a hot topic for a long time. Visual features generally include color, texture, shape and other information. Usually, the color information can be described by traditional feature extraction methods such as Hue, Saturation, Value(HSV) [28] and Histogram of Oriented Gradient (HOG) [29]; texture information can be described by the gray-level

cooccurrence matrix (GLCM) [30], autoregressive model (AR) [31] and local binary patterns (LBP) [32]; and shape information can be described by Hough [33], SIFT [34], SURF [35] and so on.

Although algorithms for extracting image features have constantly improved in recent years, this still cannot change the fact that image features are relatively low-level representations; therefore, they cannot be connected with high-level semantic information, and this problem is called the “semantic gap”. With great recent achievements [36], deep learning shows significant advantages over traditional methods, which are conducted by a series of convolutions and nonlinear activation operations.

Convolution neural networks have undergone rapid development since they were proposed, and excellent networks have constantly emerged. Currently, VGG [37], GoogleNet [38] and ResNet [39] are frequently used in ZSL. Due to the limitations of their calculation abilities, expensive professional computing resources are not possessed by many researchers, and this restricts the choice of network. Therefore, networks that are relatively small, such as VGG-16 and VGG-19, are used most frequently.

However, the use of deeper networks is of great help for improving recognition accuracy; with the recent developments of hardware technology and cost reduction approaches,

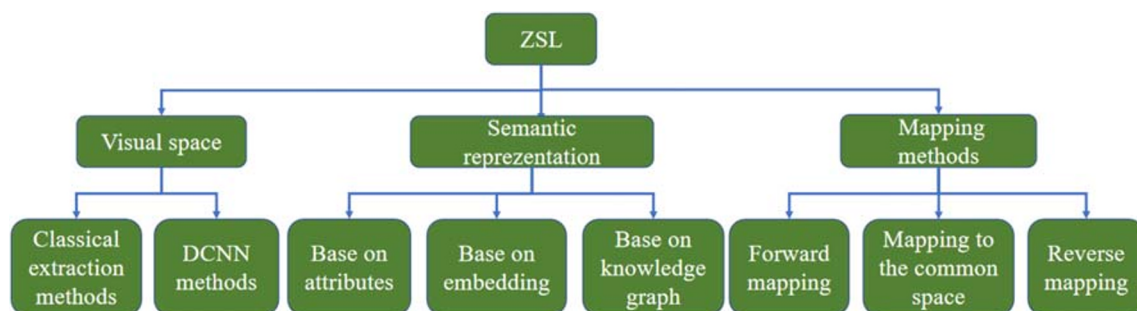


Fig. 3 Framework of ZSL

it is becoming a trend to use deeper networks. For example, GoogleNet has more layers than VGG, so the features extracted are more abstract. The “inception” module enables GoogleNet to obtain additional visual information of different scales, and the module is still constantly updated. ResNet builds a deeper network through residual learning, and it can achieve better prediction accuracy than GoogleNet and VGG. Therefore, in the cases where the hardware cost is not an obstacle, deeper networks are preferred.

For example, Chasset et al. [40] used a GRNN to train the classifier, and this greatly improved the prediction accuracy of the model. Xuesong et al. [41] proposed a deep attribute prediction (DeepAP) model based on the DAP, which can realize the mapping from low-level image features to high-level semantic attributes.

3.2 Semantic representation

ZSL can accomplish the task of identifying unseen classes, which is impossible for traditional supervised methods, because ZSL not only uses visual features for recognition but also introduces semantic attributes. The purpose of semantic attribute extraction is to construct a semantic vector, and the vector must have sufficient semantic ability as well as strong discriminant ability; unfortunately, both requirements contradict each other. Highly detailed information on objects for semantic attribute vectors is expected, including some information irrelevant to classification tasks. Nevertheless, the strong discriminant ability requires the semantic vector to pay most attention to the information that can be used to distinguish between different objects. Hascoet et al. [42] believed that different semantic representations have different influences on the accuracy of ZSL. The main semantic representation methods in ZSL include attribute-based, word vector-based and knowledge graph-based methods.

3.2.1 Semantic attribute

Semantic attributes refer to the inherent characteristics of an image or object, and they can bridge the gap between low-level features and high-level class concepts. The method of attribute generation can deal with data sparsity by transferring attribute knowledge in image and video recognition tasks.

A semantic attribute is the semantic vector of class tags manually annotated according to expert knowledge, and it is therefore a natural language attribute that can be shared among objects. It is the most common and effective method for semantic feature construction.

Attributes can usually be named, such as colors, shapes, body parts, climates, preferences, etc. Note that these definitions allow for attributes that are not directly visible but are closely related to visual information. As we all know, image features are computable, but they are not easily understood by

humans; humans associate a meaning with a given attribute name, and this allows them to create annotations directly in the form of attribute values, which can then be used by computers. Attributes can be assigned to images or classes, especially the latter, because they can form attribute annotations for new classes at the lowest possible cost; therefore, they lay the foundation for attribute-based zero-shot classification.

3.2.2 Semantic word vector

The method based on word embedding vectors automatically constructs semantic vectors by processing unsupervised text documents, and natural language processing technology is typically used. The distributed expression method includes Skip-gram [43] and CBOW [44], and this method is the most successful form of semantic representation at present because it has few dimensions and the closest distance between semantically similar words.

In recent years, the word vector method has been widely used in ZSL. The user-defined semantic attribute space is predefined according to human experts or conceptual ontology, and each dimension of the space has a specific semantic meaning. The semantic word vector space is trained by the NLP model (Wikipedia) and other language knowledge bases. Although the relative positions of different visual concepts have semantic meanings, such as the fact that a horse is more like a dog than a tree, each dimension of the space has no specific semantic meaning.

Unlike unsupervised language models, distributed word vector representations help to model syntactic and semantic rules in languages and support vector-oriented reasoning. For example, few ontology descriptions of the target domain and the general embedded word space are used for generalization [45], and then the parameters of the zero-shot semantic analyzer are refined by using user feedback to improve the online performance of the model, as seen in similar research [46, 47].

3.2.3 Knowledge graph

Methods based on knowledge graphs are different from the above two types in that they directly use class similarity to construct semantic vectors, and each element of each vector represents the class similarity. WordNet is a wide-ranging English lexical semantic web through which the minimum common upper subset of similarities for all unseen classes and seen classes can be extracted. Then, the path length is used to calculate the similarities between each class and all other classes in the upper subset. Finally, these similarities are used to build semantic vectors with hierarchical characteristics. Recently, Gao et al. [48] proposed an end-to-end ZSL framework based on a knowledge graph; the framework is a two-stream convolutional network (TS-GCN) that makes full use of a knowledge graph to realize zero-shot recognition of actions.

Currently, in addition to the above three methods of semantic attribute extraction, researchers continue to try new approaches. For example, Karessli et al. [49] proposed a method that relies on the human attention mechanism as auxiliary information and used three new embeddings to learn the compatibility function between images and attention. The method can provide results that are comparable to those derived from expert annotations.

Based on the fact that ZSL model development has been relatively stable, researchers have begun to pay close attention to the innovation of fine-grained recognition models. One example is Elhoseiny et al. [50], who provided an end-to-end flexible and compact model and only coded the parts used to distinguish between categories.

3.3 Visual-semantic mapping

Visual-semantic mapping is an indispensable cornerstone for achieving ZSL, and it is a junction between image features and semantic vectors. Once a visual-semantic mapping is established, the similarity between the data of any unseen class and the seen class prototype can be calculated, and the unseen class can be predicted based on the similarity.

There are 3 ways to construct a visual-semantic mapping: 1) Forward mapping: a mapping model is built on the visual side, then image features are mapped to the semantic space, and the recognition of unseen classes is completed in the semantic space. 2) Common space mapping: The mapping model is built on both sides. Image features and semantic vectors are mapped to the common space at the same time, and unseen classes are recognized in the public space. 3) Reverse mapping: a mapping model is built on the semantic side, through which the semantic vectors are mapped to the visual space, and the recognition of

unseen classes in the visual space is finished. The details of these three approaches are shown in Fig. 4.

In the above figure, f_v stands for the visual-space mapping function, and f_s represents the semantic space mapping function. The forward mapping function has been used since 2009, and it is a mainstream mapping method for ZSL. For example, a method of nonlinear forward mapping was conducted by the authors in [51]; their method automatically discovers and captures the attribute relationship of an object through a uniform probability model. In 2013, Akata et al. [52] first proposed common space mapping by using the bilinear equation, and this method had good judgment and generalization abilities. In 2015, reverse mapping was first adopted, and it can be used to construct virtual training samples for unseen classes, thus allowing for direct supervised classification learning in the image feature space. For example, Bucher et al. [53] employed the DAE/GAN model to map an unseen class attribute vector to the visual space, and they used SVM for supervised classification. Their method converted the problem of ZSL into a supervised learning problem.

According to the analysis of a large number of existing works, it has been found that although many forward mapping methods have been adopted, the utilization rate of reverse mapping is increasing each year because deep learning technology can extract good image feature structures that have compact distributions of similar data and strong discrimination ability; therefore, reverse mapping methods will continue to exhibit rapid development.

4 Mainstream ZSL models

In recent years, a large number of network architectures for ZSL have originated from two classical network models, DAP

Fig. 4 Mapping diagram. (a) Forward mapping; (b) Common space mapping; (c) Reverse mapping

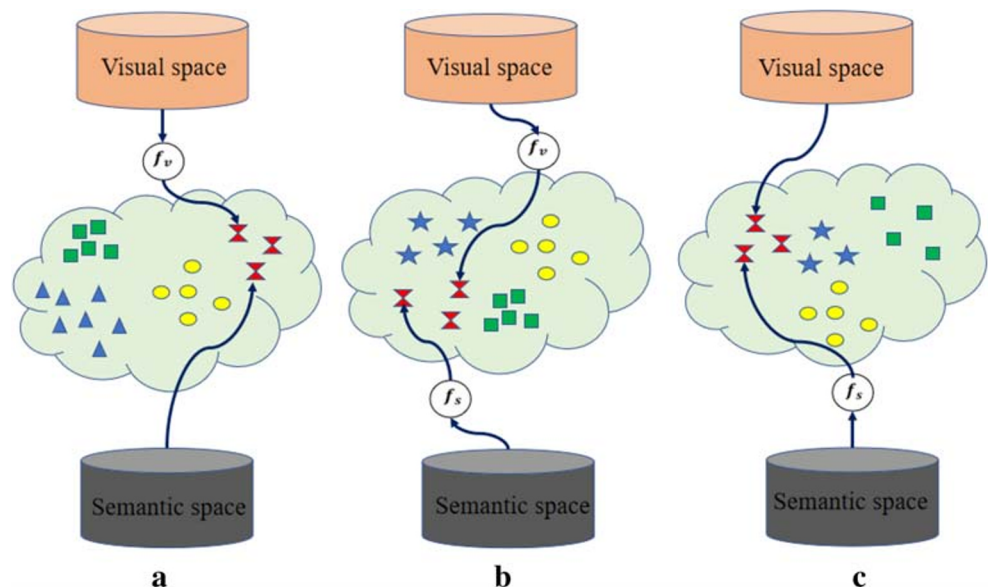
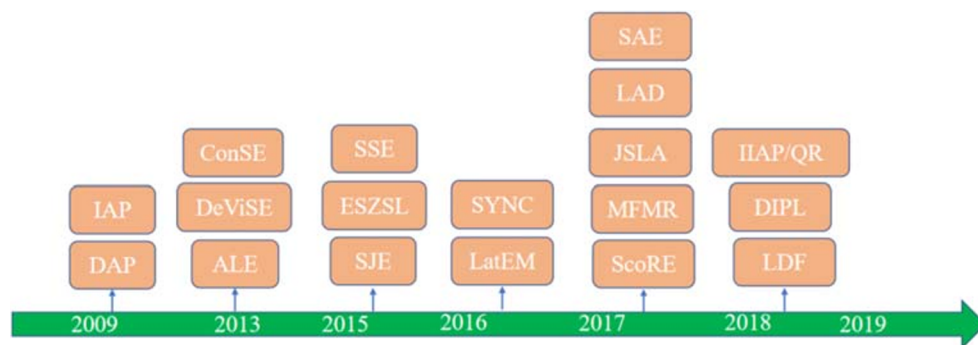


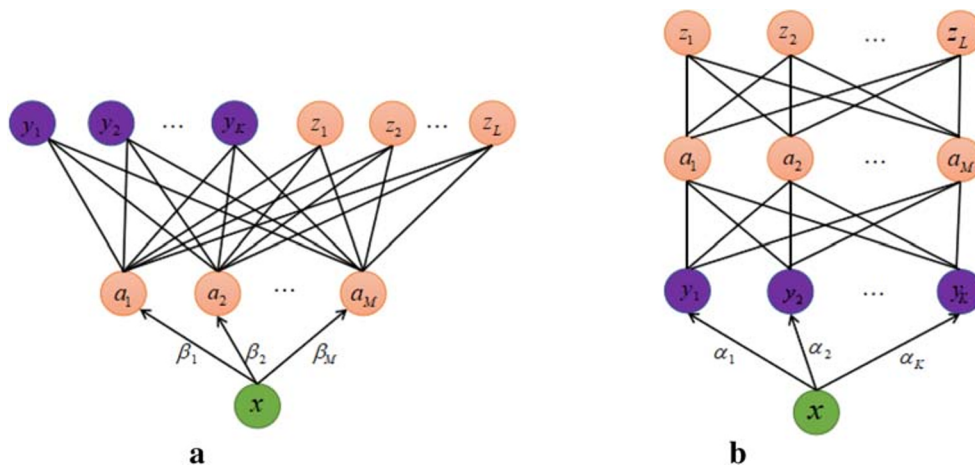
Fig. 5 Development of zero-shot learning



and IAP. Therefore, a chronological list of famous architectures is presented in Fig. 5.

- 1) **DAP**: It is a typical example of an inductive-type ZSL method. This model uses training data (such as pixels) to directly learn the mapping from image features to public attributes, and then the public attributes are used to match the class of the image. As shown in Fig. 6(a), the model can be understood as a three-layer model. The first layer is the input layer; the second layer represents a feature space of p -dimension (each dimension represents a feature); the third layer is the output layer, which shows the judgment result of the class. Between the first and second layers, there is an area used to judge whether an image is in line with the features corresponding to the p -dimensional visual space. Between the second and third layers, there is a knowledge base, which is used to store the correspondence between the p -dimensional feature space and the output “ y ”. A pattern is found in that DAP is good at judging attributes, but it performs poorly in predicting categories. Moreover, it is unable to take advantage of new samples to improve the classifier’s performance; for example, it cannot make use of additional attribute information such as WordNet.
- 2) **IAP**: IAP is obviously different from DAP, and the IAP model constructs the attribute classifier by combining the probabilities of all the associated seen classes, as shown in Fig. 6(b). During the training steps, the probability of all the training classes can be estimated by learning the probabilities of the multiclass classifier, and then the classification method is the same as that of the DAP model.
- 3) **IIAP/QR** [54]: This model is an incremental ZSL framework based on IAP. First, a fast incremental classifier based on linear discriminant analysis and QR updating is developed, and this classifier can solve the problems of few samples and nonuniform samples that often occur in incremental learning. Then, through the probabilistic reasoning of the class attribute layer and the zero-shot attribute classification layer, the model can effectively update itself.
- 4) **ALE** [55]: What makes the ALE model special is that instead of studying how to extract the appropriate feature from an image, it works on mapping a category to the appropriate Euclidean attribute space. The ALE model learns a function “ f ” that measures the matching rate between the image and attribute vector. The ALE model uses semantic information (such as the attribute)

Fig. 6 Model sketch. (a) DAP; (b) IAP



to embed each label in the p -dimensional semantic space, and the model directly learns the mapping between the feature space and the semantic vector; therefore, the ALE training process is guided by the category directly predicted. When testing, the mapping function can ensure that the points mapped in the semantic space are far away from the points corresponding to those categories that have appeared in the training process and ensure that the semantic vector corresponding to the correct zero sample class is close. It is emphasized that the model performance can be improved with additional category assistance information, which can be converted to prior knowledge for other kinds of classes if needed.

- 5) DeViSE [56]: An embedding vector from language learning is used to conduct regularization, which is used to map labels to the target vector representation. The softmax prediction layer of this model is replaced by a projection layer and a similarity measurement layer, which are trained to predict these vectors for each image. Regarding the loss function, a combination of dot-product similarity [57] and hinge rank loss [58] is used, and the model is trained by synchronous stochastic gradient descent on a distributed computing platform [59]; therefore, it is faster than the linear search method [60].
- 6) ConSE [61]: This model is completely different from the DeViSE model in that it maintains the integrity of the softmax layer of the convolution network and does not require further training of the neural networks. For a given test image, ConSE can run Krizhevsky et al.'s [62] classifier and consider the top T predictions of the model. Then, the convex combination of T semantic embedding vectors in the semantic space is calculated, and this step defines the deterministic transformation from the output of the softmax classifier to the embedding space.
- 7) SJE [63]: In this model, the ZSL of an image is carried out by determining the labels that produce the highest correlation compatibility score. To solve the problem of fine-grained ZSL, a compatibility function is constructed to obtain higher scores for matched embeddings than for mismatched embeddings. A new weakly-supervised word2vec variant was proposed, and it can be combined with other outputs to improve fine-grained recognition accuracy.
- 8) ESZSL [64]: This model is based on a general framework. The framework models the relationships among features, attributes, and classes as two linear networks, where the weights at the top are not learned but given by the environment. This method provides a strategy for learning generalization errors and converting them into a domain adaptation method.
- 9) LatEM [65]: This model also focuses on the bilinear framework and provides a new latent embedding model for learning the compatibility function between image and class embeddings. This method extends the bilinear compatibility model (ESZSL) by adding latent variables. Instead of learning about a single bilinear mapping, this model learns a set of mappings. The model is trained through a sort-based target function, which punishes incorrect orderings of the real classes for a given image.
- 10) SCoRE [66]: Some methods adopt independent learning semantics (Deep-RIS), while others supervise a subspace (Deep-RULE) by training classes. The former type, however, can constrain the entire space but lacks the ability to perform semantic association. Although the latter has the ability of semantic correlation, part of the semantic space is unsupervised, and this aggravates the problem of semantic domain shift. Therefore, the author utilized the complementarity of both types in a new convolution neural network (CNN) framework, which used semantic constraints for recognition, and the transfer capability of the CNN was supported by a hidden semantic layer and a semantic code.
- 11) MFMR [67]: This model is a new framework for multiple regularized projection based on matrix trifactorization. Specifically, under the guidance of semantic embedding and the class label matrix, the semantic embedding projection is learned by decomposition of the visual feature matrix. By introducing the manifold regularization learning projection into the visual data and semantic embedding, the geometric manifold structure can be captured effectively, and the projection domain shift problem can be avoided.
- 12) SAE [68]: It is also a solution for the domain shift problem. The attribute layer generated by the input layer of the model is transformed so that it has the function of restoring itself to the original input layer; furthermore, the mapping layer must have the limitation of a specific semantic meaning, thereby ensuring that the mapping function retains as much information as possible about the original input layer.
- 13) JSLA [69]: Most of the previous attribute learning methods focused on named semantic attributes but ignored the meaning of latent shareable visual attributes; therefore, a new dictionary learning model was proposed, and it decomposes the dictionary space into three parts: semantic attributes, latent distinguishing attributes and latent background attributes. Then, this joint attribute learning model is extended by a multitask transfer learning framework to solve the problem of unsupervised domain adaptation.
- 14) LAD [70]: This model also pays attention to the influence of the correlations between attributes on the recognition effect. A potential attribute space learning method was proposed, and it was not only discriminative but

also had semantic restrictions. Specifically, the model uses the dictionary learning framework to associate latent attribute spaces with attribute spaces and similar spaces.

- 15) SYNC [71]: This model connects seen and unseen classes by adding a set of phantom object classes and derives new embeddings through convex combinations of these phantom classes, through which the alignment between the semantic space and visual space is achieved.
- 16) LDF [72]: Many works focus on learning the correct mapping of visual and semantic spaces but ignore the importance of learning discriminant representations. An end-to-end network was developed, and this network can automatically discover the recognition region through the zoom network.
- 17) DIPL [73]: Learning the projection function between the feature space and semantic space is regarded as the key to ZSL, and this function can effectively resist the huge domain gap between seen and unseen classes. The model has two new components: the first is the introduction of the domain-invariant feature self-refactoring task in the seen/unseen class data. Second, the learning of a mapping can further align the two domains, and a super-class is formed in the semantic space to realize semantic structure sharing between the seen and unseen classes.
- 18) SSE [74]: Based on the use of each source/target data pair as a seen class mixing ratio, if the two samples belong to the same unseen classes, the mixed mode must be similar; this inspired studies on learning embedded source/target functions. These functions map arbitrary source/target domain data to the same semantic space; therefore, a maximum margin framework was developed for learning the similarity function.

After the introductions of the above models, some conclusions are drawn as follows:

The similarity between the DAP and IAP models is that a support vector machine (SVM) is used to learn embeddings, and the Bayesian formula is adopted for recognition. However, the difference is that the DAP is good at identifying attributes, while the IAP model is skilled at recognizing classes. Among these models, the IAP/QR model is an extended of IAP, while ALE is an extension of the DAP model.

Four of these models implement different strategies for the projection domain shift problem. The MFMR model seeks solutions from the perspective of multiple regularization projections of matrix trifactORIZATION. The SAE model has a specific semantic meaning from the mapping layer, and the SYNC model is executed by adding a set of phantom objects to connect to the seen and unseen classes. However, DIPL employs the domain-invariant projection learning method. Regardless, the ultimate goal of each method is to align the

semantic space with the manifold structure of the visual space to address the projection domain shift problem.

Three of the aforementioned models are related to embedding. The similarity between the DeVISE and ConSE models is that they both assign the embedded image space to the most recent class; however, attributes captured by the LatEM model correspond to different latent variable mappings. This latter method performs significantly better than the other two for fine-grained object recognition, and it can produce highly interpretable results.

The ALE, DeVISE, and SSE models are similarity measurement methods. For numeric data, the similarity of two objects refers to the degree of their adjacency to each other in Euclidean space. For classification data, the similarity of two objects is related to the number of attributes with the same value. The most common distance estimation methods are Euclidean distance and non-Euclidean distance. Euclidean distance is measured based on the locations of points in space, while non-Euclidean distance measurement is performed based on the attributes of points. Therefore, the choice of the appropriate measurement method should depend on the requirements of the target task.

The models centered around dictionary learning include JSLA and LAD, which have strong commonality, simplicity and efficiency. Such methods are no longer limited to comprehensive sparse models, analytical models, blind dictionary models and so on. Simplicity measurement expands the comprehensive sparse model and makes the dictionary learning model diverse.

ZSL based on public mapping usually involves a bilinear framework model, such as ESZSL or LatEM. During the process of training these models, the category label of the image is replaced by the embedded vector with semantic information, and the image features and semantic attributes are mapped to the same low-dimensional potential space. This method can reduce the overdependence of the trained classifier on the original data and realize the generalization of the classifier.

The SCoRE and LDF models are typical examples of end-to-end networks. Their motivation is that in existing methods, the extraction of low-level features and the construction of the embedded space are handled separately; as a result, it is impossible to learn many latent distinguishing features in the visual and semantic spaces; however, they have a key impact on the ZSL effect. Therefore, it makes sense to build end-to-end networks that can learn about latent distinguishing features.

5 Generalized zero-shot learning (GZSL)

Conventional ZSL refers to cases where only seen classes are used in the training stage, and unseen classes are used in the testing stage. Although this assumption can greatly simplify the recognition task, it is not realistic in actual applications;

after all, both seen and unseen classes may appear during the test phase, and this is called GZSL.

First, it is necessary to distinguish whether the object to be recognized belongs to the seen classes or unseen classes and then determine the specific membership class. In 2013, Socher [75] introduced GZSL from the perspective of transfer learning. This specific implementation is executed by first using outlier detection in the semantic space, and then two separate recognition models are developed. In 2016, Chao [76] first proposed the concept of GZSL, and Song [77] first used the transductive model to explore the setting problem of GZSL, effectively solving the problem of classification bias. A low-dimensional embedded method was developed by the authors of reference [78], and it alleviates the semantic gap caused by mapping high-dimensional visual features to the semantic space. A new deep calibration network (DCN) [79] was proposed that can calibrate the confidence of the source class and the uncertainty of the target class simultaneously. This method mapped the visual feature of an image and the semantic representation of a class prototype into a common embedded space, thereby ensuring the maximum compatibility of the data from the source and target classes. Arora [80] built an architecture based on variational autoencoders. The key to this architecture is the feedback-driven mechanism, in which a discriminator maps the generated examples to corresponding class attribute vectors. Then, an improved generator was obtained that can alleviate the bias of predicting seen classes in GZSL. It is worth mentioning that Shafin [27] proposed a general strategy to solve the traditional ZSL, GZSL and few-shot learning (FSL) problems. The method was based on a new kind of class-adaptive principal direction (CAPD) concept, which allowed for multiple image features embedded in a semantic space. For this reason, efficient and adaptive information transfers between seen classes and unseen classes can be achieved.

Although some achievements have been made in exploring the problem of adding new classes to deep neural networks

while maintaining the ability of the model to recognize original classes, lifelong learning and the gradual addition of new classes remain unsolved problems.

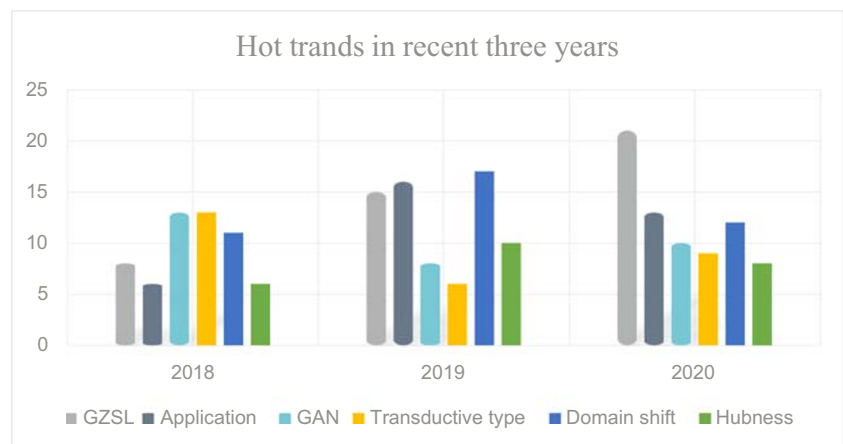
6 Analysis of relevant papers

6.1 Paper collection and analysis

Everyone has limitations in terms of judging the research hotspots in ZSL. If a scientific and fair judgment is made, at least three years of recent, closely related papers need to be collected to obtain insight into current hot issues and predict future research directions. The specific operations for collecting papers are as follows: based on some keywords, such as zero-shot learning, visual-semantic mapping, semantic representation, semantic attribute, generative adversarial network (GAN) and so on, 213 papers in the last three years are collected. The studies involve many aspects of ZSL, six of which are hot: GZSL research, practical applications of ZSL, exploration based on GANs, transductive zero-shot learning, exploration of the domain drift problem, hubness and so on, as shown in Fig. 7.

As shown in Fig. 7, studies on GZSL [81–85] have increased steadily in the last three years, and this is not surprising because unseen classes can only be used in the testing stage in traditional ZSL, which is often unrealistic. Therefore, the exploration of GZSL is market-oriented and had an absolute quantity advantage in the first half of 2020. Second, there are many articles [86–93] on practical applications of ZSL. After all, the traditional ZSL method only stops in the laboratory stage, and many restrictions are too harsh; however, there are many complex and uncontrollable factors in actual scenarios, and this further stimulates the desire for research. Third, the number of methods based on the GAN [94–100] showed a decline in 2019 but displayed a significant growth in momentum during the first half of 2020 due to the

Fig. 7 Distribution of related research in the last three years



GAN's ability to generate new samples. It is worth noting that it is able to not only generate diverse samples within classes but also generate samples with significant differences between classes. There is a similar development law between transductive ZSL [101–103] and the GAN. Finally, the projection domain drift and hubness problems are recognized as two major challenges in ZSL. Some existing methods [104–107] alleviate these problems to a certain extent, but there is still a lack of effective solutions.

6.2 Challenges and the future

ZSL empowers a machine with the ability to make inferences and judgments based on “words and phrases” like humans do. At present, however, ZSL is not as hot as deep learning algorithms. The main reason for this is that several technical problems have not been effectively solved.

6.2.1 Projection domain shift

The effect of ZSL depends on the information of similar modes. When a semantic vector model based on attributes is constructed, the same attribute belonging to different categories may have great differences in visual features. For example, a rabbit's tail and a mouse's tail both have the same tail attributes, but the visual characteristics of the two tails are very different from each other; therefore, it is difficult for the model trained by the rabbit to recognize the mouse, and this effect is called the attribute shift phenomenon. If the classes of the training set and the test set are too different, the underlying data distribution of these classes may also be different. For example, if the training set is all birds while the test set is all electronic components, then it is too difficult for ZSL methods to analyze the mapping relationship between the two, and this is called the “strong bias” problem with attribute drift, which significantly reduces the recognition performance of ZSL. Reconstructing the samples mapped to the semantic space is feasible [75] and doing so can result in much retained information. For the domain shift problem, a joint generative model called Zero-VAE-GAN [108] was developed to generate high-quality unseen features, through which ZSL was converted to traditional supervised learning. Although the projection domain shift can be ameliorated [109–111], it has never been effectively solved.

6.2.2 Hubness

In a high-dimensional space, some points become the nearest neighbors of most points, and this is an inherent property of high-dimensional spaces. The K-NN algorithm is typically used in the final classification of the ZSL model, and the hubness problem [100] seriously affects the classification results. It is assumed that when the model is built by ridge

regression, the visual-semantic mapping method needs to be changed to reverse mapping, which can reduce the hubness problem. When a generative model (such as a self-encoder or GAN, etc.) is used to generate test set samples, ZSL can be converted into traditional supervised classification [112, 113], thereby avoiding the KNN operation to eliminate the hubness problem.

6.2.3 Reducing the cost of ZSL

ZSL lacks sufficient professional definition and description; nevertheless, this process is more effective when done by humans than by machines. However, there is still insufficient professional assistance at present, and the development of natural language processing (NLP) itself is not enough to meet the needs of ZSL, making the overall process relatively slow.

Exploring how to use existing text content on the Internet (such as Wikipedia) is very important for implementing improved algorithms because text content on the Internet is easily available; therefore, it can greatly reduce the work cost of ZSL so that ZSL can be extended to additional applications.

6.2.4 Semantic representation

The paper shows that attributes are better than word vectors, but word vectors are easier to obtain than attributes. Therefore, an ideal semantic vector space is one that includes the advantages of both, and such a space is of great help for solving the technical problems of ZSL.

6.2.5 Application of reinforcement learning

Some studies combine ZSL with reinforcement learning, that is, they introduce the target design of ZSL into the model of reinforcement learning so that the agent can perform tasks using knowledge that has not been taught in the training stage.

6.2.6 Unified model

FSL, one-shot learning and ZSL are also used to solve the problem of identifying scarce samples. FSL and one-shot learning adopt completely different strategies from that of ZSL. However, ZSL can be regarded as the extreme case of FSL, and exploring a unified model by combining ZSL with one-shot learning and FSL is a future research direction.

In addition to the issues mentioned above, there are some other problems, such as semantic loss in ZSL and samples collapsing to a point through mapping. If these constraints are not addressed, the ZSL model will be robbed of opportunities by algorithms inferior to it, even if it has the potential to go from its current level to excellent studies.

7 Conclusions

This paper summarizes ZSL methods according to 113 references in the following aspects: the evolution of network architectures, key technologies, typical models, GZSL, hot spot analysis and future challenges are discussed in detail. First, the difficulty of ZSL is analyzed from the perspective of the evolution of network architectures. Second, the advantages of the visual feature extraction method based on deep learning are interpreted, and the advantages and disadvantages of semantic representation methods based on attributes, word vectors and knowledge maps are summarized; visual-semantic mapping, including forward mapping, reverse mapping and public mapping, is discussed. Eighteen typical ZSL models are listed, and it is found that the end-to-end networks have unique advantages. Finally, relevant articles (213) from the last three years are collected to complete Fig. 7, through which current hot spots and future research directions in this field are examined.

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Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest.

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